Units, pressure, temperature *(Ch. 1)*

$R_u = 8.314\,\text{kJ/(kmol·K)}$ $g = 9.807\,\text{m/s}^2$ $\sigma = 5.67\times10^{-8}\,\text{W/(m}^2\text{·K}^4\text{)}$
$1\,\text{atm} = 101.325\,\text{kPa} = 760\,\text{mmHg}$ $1\,\text{bar} = 100\,\text{kPa}$ $1\,\text{L} = 10^{-3}\,\text{m}^3$
$\mathbf{1}\,\text{kPa·m}^3 = \mathbf{1}\,\text{kJ}$ $\mathbf{1}\,\text{kJ/kg} = \mathbf{1000}\,\text{m}^2/\text{s}^2$ $1\,\text{kW} = 1\,\text{kJ/s}$ *(divide $V^2/2$ by 1000 to get kJ/kg)*
$T(\text{K}) = T(^\circ\text{C}) + 273.15$ $T(^\circ\text{F}) = 1.8\,T(^\circ\text{C}) + 32$ $T(\text{R}) = 1.8\,T(\text{K})$
$\Delta T(\text{K}) = \Delta T(^\circ\text{C})$

$$v = \frac{\mathcal{V}}{m} = \frac{1}{\rho} \quad \text{SG} = \frac{\rho}{\rho_{H_2O}},\ \rho_{H_2O} = 1000\,\text{kg/m}^3,\ \rho_{Hg} = 13\,600\,\text{kg/m}^3$$

$$P = \frac{F}{A} \quad P_{abs} = P_{atm} + P_{gage} = P_{atm} - P_{vac}$$ *(tables and gas laws need absolute P and T)*

$\Delta P = \rho g\,\Delta z$ $P = P_{atm} + \rho g h$ $\Delta P = -\int \rho g\,\mathrm{d}z$ *(if ρ varies)*

Manometer walk. Start at a known pressure. Add $\rho g h$ for each fluid layer on the way *down*, subtract it on the way *up*. Same level in the same connected fluid means same pressure: $P_1 + \sum_{\text{down}} \rho g h - \sum_{\text{up}} \rho g h = P_2$

Barometer: $P_{atm} = \rho_{Hg}\, g\, h$ Pascal: $\dfrac{F_1}{A_1} = \dfrac{F_2}{A_2}$

State postulate: two independent intensive properties fix the state of a simple compressible system. Closed = fixed mass; control volume = mass crosses the boundary; isolated = neither mass nor energy crosses. Steady = nothing in the CV changes with time.

Energy, work, heat transfer *(Ch. 2)*

$$E = U + \tfrac{1}{2}mV^2 + mgz \quad e = u + \frac{V^2}{2} + gz \quad \dot{E} = \dot{m}\,e$$

$$e_{mech} = \frac{P}{\rho} + \frac{V^2}{2} + gz$$

$$\Delta\dot{E}_{mech} = \dot{m}\left[\frac{P_2 - P_1}{\rho} + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)\right]$$

Pump: $\eta_{pump\text{-}motor} = \dfrac{\Delta\dot{E}_{mech,fluid}}{\dot{W}_{elect}}$ $\Delta P_{pump} = \dfrac{\Delta\dot{E}_{mech}}{\dot{\mathcal{V}}}$
(turbine-generator: invert the ratio)

Signs: $Q > 0$ into the system, $W > 0$ out of it. The slides book $Q_{in}, Q_{out}, W_{in}, W_{out}$ as positive magnitudes: $Q = Q_{in} - Q_{out}$, $W = W_{out} - W_{in}$. Heat is driven by ΔT; any other energy crossing the boundary is work. Both are path functions (δQ, δW).

$$W_b = \int_1^2 P\,\mathrm{d}\mathcal{V}$$ *(area under the P-$\mathcal{V}$ path; >0 expansion)* const. P: $P(\mathcal{V}_2 - \mathcal{V}_1)$ const. $\mathcal{V}$: 0

Polytropic $P\mathcal{V}^n = C$: $W_b = \dfrac{P_2\mathcal{V}_2 - P_1\mathcal{V}_1}{1-n} = \dfrac{mR(T_2 - T_1)}{1-n}$ $(n \neq 1)$

$n = 1$: $W_b = P_1\mathcal{V}_1 \ln\dfrac{\mathcal{V}_2}{\mathcal{V}_1} = mRT\ln\dfrac{\mathcal{V}_2}{\mathcal{V}_1}$ *(isothermal ideal gas)*

$W_{sh} = 2\pi n\,\text{T}$, $\dot{W}_{sh} = 2\pi\dot{n}\,\text{T}$ *(n in revolutions, torque T)*
$W_e = (\text{volts})(\text{amps})\,\Delta t$

$$\dot{Q}_{cond} = -kA\frac{\mathrm{d}T}{\mathrm{d}x} = kA\frac{T_1 - T_2}{L} \quad \dot{Q}_{conv} = hA(T_s - T_\infty)$$

$\dot{Q}_{rad} = \varepsilon\sigma A\left(T_s^4 - T_{surr}^4\right)$ *(T in K)* $\dot{q}_{absorbed} = \alpha\,\dot{q}_{incident}$ $Q = \dot{Q}\,\Delta t$, $q = \dfrac{Q}{m}$

Cooling body: $mc\dfrac{\mathrm{d}T}{\mathrm{d}t} = -\dot{Q}$, so the rate goes as $A/(mc)$. *Sphere has the least area for a given volume.*

Properties of pure substances *(Ch. 3)*

Five states, in heating order. Compare with $T_{sat}(P)$ or $P_{sat}(T)$:

State	You are here if	So you can say
1. Compressed (subcooled) liquid *A-7*	$T < T_{sat}(P)$, $P > P_{sat}(T)$, $v < v_f$	$v \approx v_f(T)$, $u \approx u_f(T)$, $h \approx h_f(T) + v_f(T)\,[P - P_{sat}(T)]$. Read at T, not at P.
2. Saturated liquid *A-4, A-5*	$T = T_{sat}$, $x = 0$	$y = y_f$; T or P alone fixes it.
3. Saturated mixture *A-4, A-5*	$T = T_{sat}$, $v_f < v < v_g$, $0 < x < 1$	$y = y_f + x\,y_{fg}$. T and P are tied together, so the second property has to be x, v, u or h.
4. Saturated vapour *A-4, A-5*	$T = T_{sat}$, $x = 1$	$y = y_g$; T or P alone fixes it.
5. Superheated vapour *A-6*	$T > T_{sat}(P)$, $P < P_{sat}(T)$, $v > v_g$	T and P are independent, read both. Ideal gas only at low density (low P, high T).

y is any of v, u, h. Quality x exists only in states 2 to 4.

$$x = \frac{m_g}{m_f + m_g} = \frac{y - y_f}{y_{fg}} \quad y_{fg} = y_g - y_f \quad \mathcal{V} = mv$$ interpolate:

$$y = y_0 + \frac{y_1 - y_0}{x_1 - x_0}(x - x_0)$$

$\boldsymbol{h = u + Pv}$ $H = U + P\mathcal{V}$ ideal gas: $h = u + RT$, $u = u(T)$, $h = h(T)$

$Pv = RT$ $P\mathcal{V} = mRT = NR_uT$ $R = \dfrac{R_u}{M}$ $\rho = \dfrac{P}{RT}$ $\dfrac{P_1\mathcal{V}_1}{T_1} = \dfrac{P_2\mathcal{V}_2}{T_2}$ *(fixed mass)*

$\mathrm{d}u = c_v\,\mathrm{d}T$ $\mathrm{d}h = c_p\,\mathrm{d}T$ $c_p - c_v = R$ $k = c_p/c_v$ $c_p = \dfrac{kR}{k-1}$, $c_v = \dfrac{R}{k-1}$

Δu, Δh of an ideal gas, best first: (1) $u(T)$, $h(T)$ from Table A-17; (2) $\int c(T)\,\mathrm{d}T$; (3) $c_{avg}\,\Delta T$ if ΔT is small. *If the test hands out A-17, use it, also backwards to get T from h.*

Air: $R = 0.2870$, $c_p = 1.005$, $c_v = 0.718\,\text{kJ/(kg·K)}$, $k = 1.4$, $M = 28.97\,\text{kg/kmol}$ water: $c = 4.18\,\text{kJ/(kg·K)}$

Polytropic, ideal gas: $\dfrac{T_2}{T_1} = \left(\dfrac{\mathcal{V}_1}{\mathcal{V}_2}\right)^{n-1} = \left(\dfrac{P_2}{P_1}\right)^{\frac{n-1}{n}}$ $n = 0$ *isobaric*, 1 *isothermal*, k *isentropic*, ∞ *isochoric*

Incompressible: $c_p = c_v = c$, $\Delta u = c\,\Delta T$, $\Delta h = c\,\Delta T + v\,\Delta P$ *(heater: $\Delta P = 0$; pump: $\Delta T = 0$, $\Delta h = v\,\Delta P$)*

Sketches. T-v: an isobar is steep in the liquid, flat across the dome, rising in the vapour; higher P sits higher.

First law, closed systems *(Ch. 4)*

$E_{in} - E_{out} = \Delta E_{sys}$ $Q - W = \Delta U + \Delta KE + \Delta PE$ stationary: $Q - W = \Delta U$, $q - w = \Delta u$

Split the work, $W = W_b + W_{other}$ *(shaft, electrical, stirring)*:

const. P: $\boldsymbol{Q - W_{other} = \Delta H = m(h_2 - h_1)}$ *(because $\Delta U + W_b = \Delta H$)*

const. $\mathcal{V}$: $Q - W_{other} = \Delta U = m(u_2 - u_1)$

Ideal gas, constant c, any process: $\Delta U = mc_v\,\Delta T$, $\Delta H = mc_p\,\Delta T$. *For steam use the tables.*

Adiabatic: $-W = \Delta U$ Isothermal *ideal gas* only: $\Delta U = 0$, so $Q = W$

Insulated rigid container, several bodies reaching one T_f: $\sum m_i c_i\,(T_f - T_i) = 0$

Cycle: $\Delta U = 0$, $W_{net} = Q_{in} - Q_{out}$ *(magnitudes)* $= \oint P\,\mathrm{d}\mathcal{V}$ *(enclosed area, clockwise positive)* $\eta_{th} = \dfrac{W_{net}}{Q_{in}} = 1 - \dfrac{Q_{out}}{Q_{in}}$

Rate form: $\dot{Q} - \dot{W} = \dfrac{\mathrm{d}E_{sys}}{\mathrm{d}t}$ *($= 0$ when steady)*

Control volumes, steady flow *(Ch. 5)*

$\dot{\mathcal{V}} = V_{avg}A$ $\dot{m} = \rho\dot{\mathcal{V}} = \rho V_{avg}A = \dfrac{V_{avg}A}{v}$ $\dfrac{\mathrm{d}m_{CV}}{\mathrm{d}t} = \sum\dot{m}_{in} - \sum\dot{m}_{out}$

Steady: $\sum\dot{m}_{in} = \sum\dot{m}_{out}$ one stream: $\rho_1V_1A_1 = \rho_2V_2A_2$
incompressible only: $V_1A_1 = V_2A_2$

$\theta = h + \dfrac{V^2}{2} + gz$ *($h = u + Pv$ carries the flow work Pv)* $\dot{E}_{mass} = \dot{m}\,\theta$

$$\dot{Q}_{in} + \dot{W}_{in} + \sum_{in}\dot{m}\,\theta = \dot{Q}_{out} + \dot{W}_{out} + \sum_{out}\dot{m}\,\theta$$ *(steady-flow energy balance)*

One stream, per unit mass: $q - w = (h_2 - h_1) + \dfrac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)$

Nozzle, diffuser: $h_1 + \dfrac{V_1^2}{2} = h_2 + \dfrac{V_2^2}{2}$ *($\dot{Q} \approx 0$, $\dot{W} = 0$). Nozzle: V up, P down, area shrinks (subsonic). Diffuser: the reverse.*

Compressor: $\dot{W}_{in} = \dot{m}(h_2 - h_1) + \dot{m}\,q_{out}$ Turbine: $\dot{W}_{out} = \dot{m}(h_1 - h_2) - \dot{m}\,q_{out}$

Both with Δke, Δpe dropped; keep Δke if velocities are given. Several exits: $\dot{W}_{out} = \dot{m}_1h_1 - \dot{m}_2h_2 - \dot{m}_3h_3$

Throttle: $h_1 = h_2$ *(adiabatic, no work; T constant only for an ideal gas)*

Mixing chamber, heat exchanger: $\sum\dot{m}_{in}h_{in} = \sum\dot{m}_{out}h_{out}$ *($\dot{Q} = \dot{W} = 0$; in a heat exchanger each stream keeps its own $\dot{m}$)*

Every question: name the system (closed or CV), list assumptions, sketch it, carry units, convert °C to K before any gas law, circle the final answer. No marks for an answer alone.